

House Price Prediction Using Linear Regression

Artificial Intelligence & Machine Learning – Task 1

Student Details

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Task: Build & Evaluate a Linear Regression Model

Dataset: California Housing Dataset

1. Introduction

The aim of this project is to build a machine learning model that can predict house prices using the California Housing Dataset. Through this project, I learned the complete machine learning workflow, starting from loading the dataset and exploring the data to training a model and evaluating its performance.

Linear Regression was selected because it is one of the simplest and most commonly used algorithms for solving regression problems. It helps us understand how different housing features affect the predicted house price.

2. Dataset Description

The California Housing Dataset is provided by the scikit-learn library and contains information about housing areas in California. Each record includes several features related to houses and their surrounding locations.

Dataset Summary

- **Total Records:** 20,640
- **Input Features:** 8
- **Target Variable:** Median House Value (MedHouseVal)

Features Used

- Median Income (MedInc)
- House Age (HouseAge)
- Average Rooms (AveRooms)

- Average Bedrooms (AveBedrms)
- Population
- Average Occupancy (AveOccup)
- Latitude
- Longitude

After examining the dataset, I found that there were no missing values, so no additional data cleaning was required before training the model.

Figure 1: Dataset Information

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	MedHouseVal
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24	3.521
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25	3.413
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25	3.422

3. Exploratory Data Analysis (EDA)

Before building the model, I performed Exploratory Data Analysis (EDA) to better understand the dataset.

The following analyses were carried out:

- Checking the dataset shape
- Viewing dataset information
- Generating statistical summaries
- Checking for missing values
- Plotting feature distributions
- Creating a correlation heatmap
- Visualizing the relationship between median income and house prices

The heatmap showed that **Median Income (MedInc)** has the strongest positive relationship with house prices. The histograms helped understand how each feature is distributed, while the scatter plot showed that areas with higher median income generally have higher house prices.

Figure 2: Correlation Heatmap

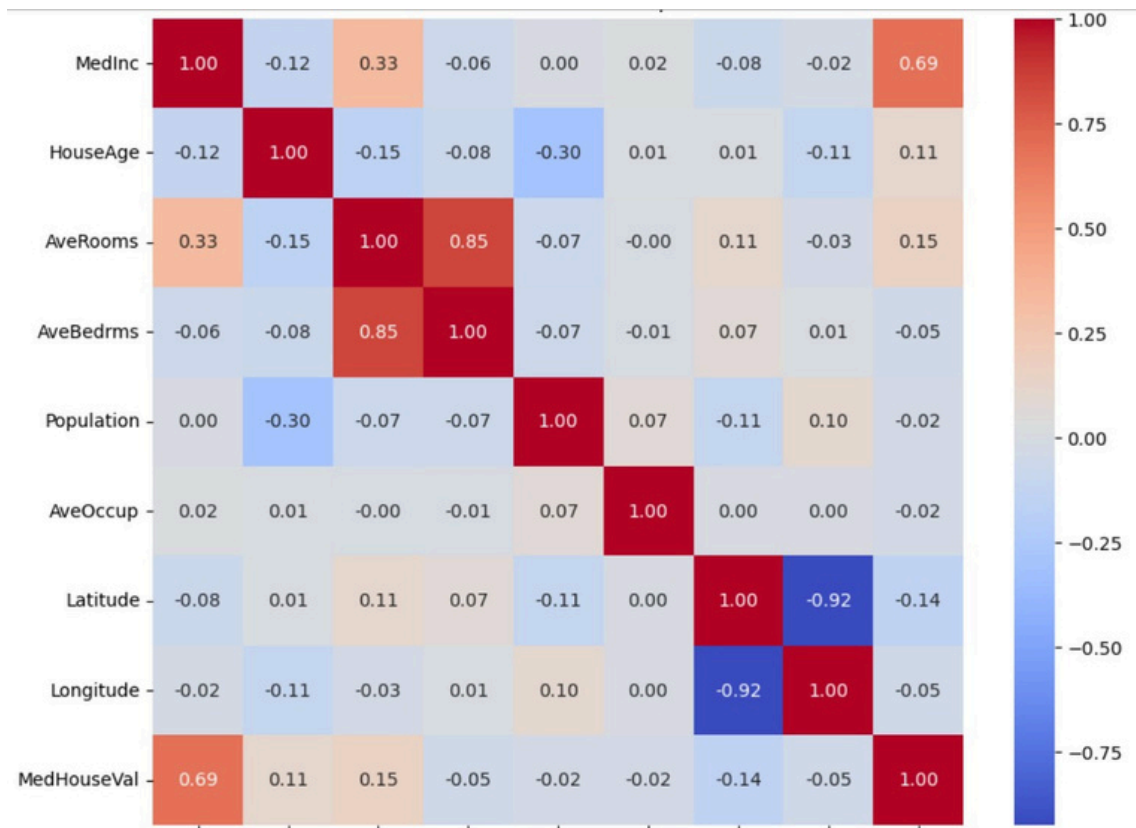
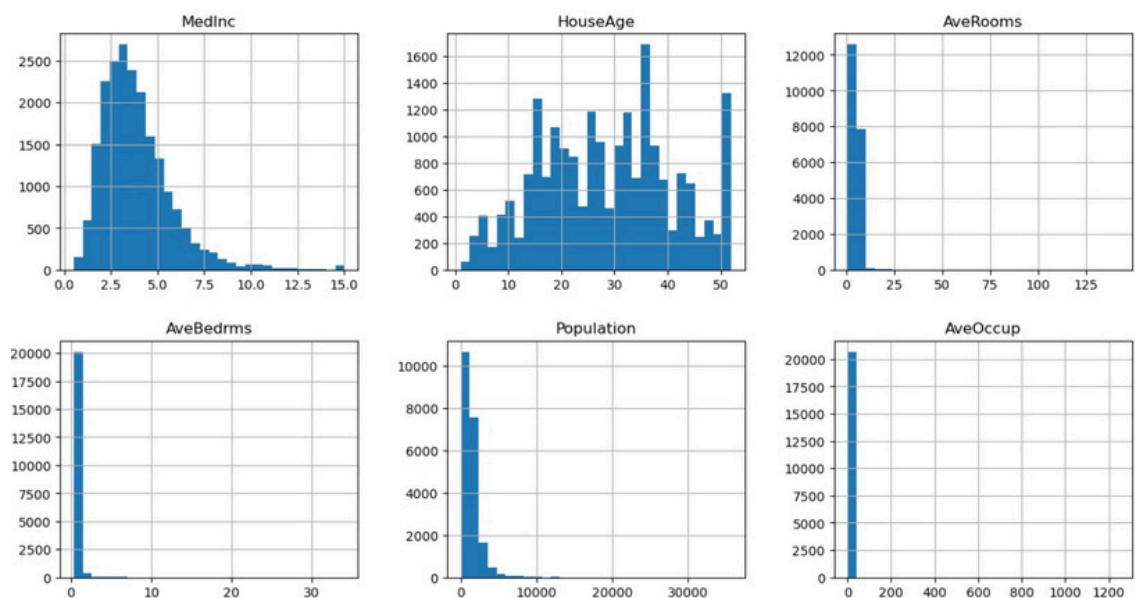


Figure 3: Feature Distribution Histograms



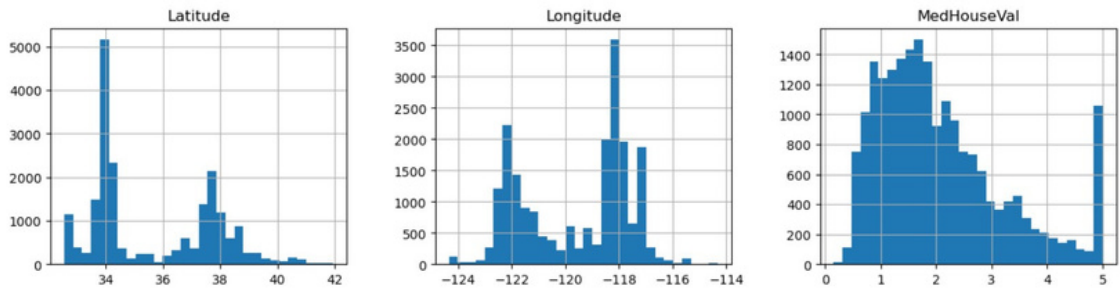


Figure 4: Median Income vs House Price Scatter Plot



4. Model Building

To build the prediction model, the dataset was divided into training and testing sets using an 80:20 ratio. The training data was used to train the Linear Regression model, while the testing data was used to evaluate its performance.

After training, the model predicted house prices for the test dataset based on the learned relationship between the input features and the target variable.

5. Model Evaluation

The performance of the model was evaluated using three commonly used regression metrics.

Evaluation Metric	Value
Mean Absolute Error (MAE)	0.533
Root Mean Squared Error (RMSE)	0.746
R ² Score	0.576

The Actual vs Predicted graph shows that the predicted values generally follow the trend of the actual house prices. Although the predictions are not perfect, the model captures the overall relationship reasonably well.

The Residual Plot indicates that most prediction errors are distributed around zero, suggesting that the model performs reasonably well for a basic Linear Regression algorithm.

Figure 5: Model Evaluation Output



Figure 6: Actual vs Predicted House Prices

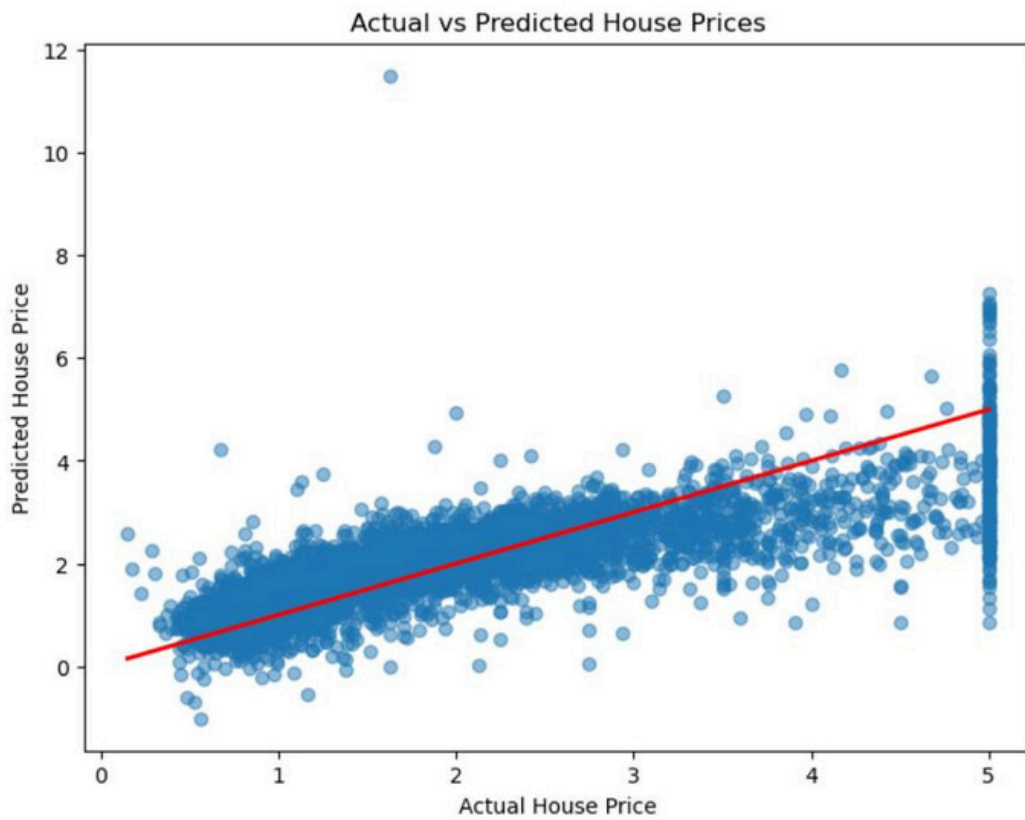
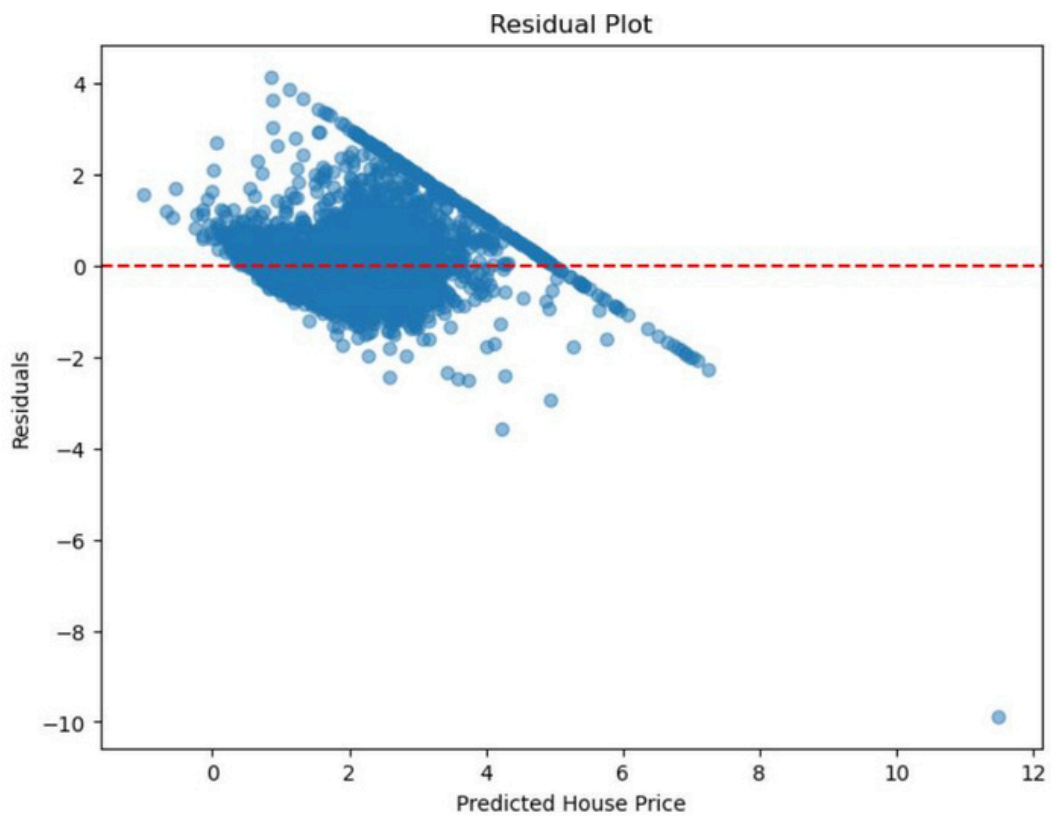


Figure 7: Residual Plot



6. Conclusion

This project successfully developed a Linear Regression model for predicting house prices using the California Housing Dataset.

Throughout this project, I gained practical experience in loading datasets, performing exploratory data analysis, visualizing data, training a machine learning model, and evaluating its performance using standard regression metrics.

The model achieved an **R² Score of 0.576**, indicating that it explains a reasonable portion of the variation in house prices. While the model provides satisfactory results, its accuracy could be improved by using more advanced machine learning algorithms such as Random Forest, Decision Tree Regression, or XGBoost, along with feature engineering and hyperparameter tuning.

Overall, this project provided valuable hands-on experience with the complete machine learning workflow and strengthened my understanding of regression-based predictive modeling.

References

1. Scikit-learn Documentation
2. California Housing Dataset Documentation
3. Python Official Documentation